



Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
MCA Department

PRACTICAL SUBMISSION RECORD- A.Y. 2024-25

Class: SYMCA	Div: B	Course Code: MCA01604	Batch: S2
Semester: III		Course Name: Data Science Laboratory	
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CO No: CO605.1		Assignment No: 05	

Title: Use a dataset with features like square footage vs house price. Implement a simple linear regression model to predict house prices. Plot the regression line and calculate MSE and R² score

Code:

```
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
import matplotlib.pyplot as plt

# Generate synthetic data for house prices
np.random.seed(42)
sqft = np.random.uniform(500, 3500, 100)
# Assume price = 150 * sqft + noise
prices = 150 * sqft + np.random.normal(0, 50000, 100)

df = pd.DataFrame({'SquareFootage': sqft, 'Price': prices})

# Split data into train and test sets
X = df[['SquareFootage']]
y = df['Price']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train linear regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict on test set
y_pred = model.predict(X_test)

# Plot results
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X_test, y_pred, color='red', linewidth=2, label='Regression Line (Test)')
plt.xlabel('Square Footage')
```

```
plt.ylabel('House Price')  
plt.title('House Price vs. Square Footage (Linear Regression)')  
plt.legend()  
plt.show()
```

```
# Calculate MSE and R^2 score  
mse = mean_squared_error(y_test, y_pred)  
r2 = r2_score(y_test, y_pred)  
  
print(f"Mean Squared Error (MSE): {mse:.2f}")  
print(f"R^2 Score: {r2:.4f}")  
print(f"Intercept: {model.intercept_:.2f}")  
print(f"Coefficient: {model.coef_[0]:.2f}")
```

Output:

```
Mean Squared Error (MSE): 1634248784.29  
R^2 Score: 0.9120  
Intercept: 10490.27  
Coefficient: 143.31
```

